

Course milestone M16: Reproducible Package and the Peer Cold Run

HONR 46400 · Evidence-Driven Research

What to submit

One document, `lastname_m16.pdf`. Everything this milestone asks for goes inside it, as sections in the order below. One artifact travels beside it, because it cannot live inside a PDF:

Submit it on Brightspace, under **Assignments** then **M16**. Brightspace carries the deadline.

Where this document has a **What this course adds** section, that section is part of the milestone and is graded, whether or not the course schedule marks this milestone.

This milestone presents **Book Milestone 11: Your reproducible package** (checkpoint, version 1) and **Book Milestone 9: Your bounded claims** (revisit, version 2 - the research note after public criticism) of the course book, EDR|AI.

- **The repaired reusable package** (shared Colab notebook, folder, or repository link): The package must be openable and runnable top to bottom in a fresh environment: notebook, data or access recipe, environment record, README, and licence. A PDF cannot be executed, so restart-and-run-all cannot be checked inside it.

What this course adds

One submission, two book artifacts

Your PDF carries two artifacts from the book's ladder. Your reproducible package is Book Milestone 11. Your research note is Book Milestone 9 written again, as version 2. Version 2 answers evidence that did not exist when you first drafted the note: real questions from readers at the Expo, and a stranger's cold run of your own package. Work both book pages, then add the course sections below.

One book instruction does not apply to you. Book Milestone 11 offers a "solo proxy" label for anyone with no external runner. You have a real peer cold run, so use it.

1. Build the inbound list

Two streams of outside evidence reached your project, and neither came from you. Gather them into one numbered table before you decide anything.

- **Expo criticism.** Pull these rows from your M15 tally and your defense revision record.
- **Your peer's cold run.** A classmate opened your package without you in the room and returned a threat-ranked list of what broke and what they had to ask you. Keep their ranking visible.

Tag every row with its source, the item in one sentence, and the part it touches: claim, evidence, writing, or package. A question you were asked five times is one row with its recurrence noted, not five rows. A reader should be able to count your rows, count your verdicts, and get the same number twice.

2. Write the adjudication record

This is the heart of the milestone, and it opens your note. Give every row a dated verdict, and exactly one of three.

1. **Accept.** The criticism was right. Say what changed and where a reader can see it.
2. **Rebut.** It does not hold. Give the evidence, not the irritation. "The reader misread the table" is the start of a reason, not the whole of one.
3. **Bound.** It holds in part, so the claim narrows. State the new boundary inside the claim sentence itself, never in a footnote.

Each verdict carries a reason in your own words, one you could defend aloud. Nothing sits undecided, and nothing is answered with silence. A claim that shrank under public testing is the system working, so say plainly that it shrank and what shrank it.

3. Revise the note, then repair the package

Revise the five-move spine wherever an accepted or bounded verdict demands it, and nowhere else. Let the note grow only where public questioning showed a reader needed more, usually to two to four pages. Order findings by how much evidence backs each: strongest first, exploratory last and clearly labeled. An AI tool may be your writing critic, never your ghostwriter. Reject every edit that upgrades a verb or quietly drops an “in this sample”.

Then rebuild the package your peer ran. Every place they had to stop and ask you something is a missing line in your documentation, so write those lines now, or log the reason the gap stays. Audit yourself against the five package sins: a hard-coded path, a missing seed, a by-hand edit, an undocumented exclusion, and stale or unversioned data. Name the one sin you found and fixed.

Redraft your AI disclosure from the full ledger rather than from memory. It now covers the poster weeks and the peer exchange as well as this revision.

4. Run it clean a second time

Any change to the package makes the previous run record stale. A number, the data, the code, the environment record, or one line of the instructions a stranger follows: each of those is a change. So after your last edit, run the package once more in a fresh environment, restart-and-run-all, following only your written instructions. Record the date, the environment, and the result. The headline number has to come back, and so does its uncertainty statement. Label the record honestly as an author rerun, and keep your peer’s run beside it as the independent evidence.

5. At the Friday studio

Two things happen at the studio itself, not before it. A peer reads your adjudication table and questions it. Then you perform the second clean run, and you submit at studio close.

What you hand in

One PDF, in this order: the inbound list, the adjudication record, the note, the updated claim-evidence table, your AI disclosure, and the clean-run record. The package cannot live inside a PDF, so share it as a link the instructor can open, holding the README, the seeded notebook, your data or its access recipe, the environment record, and the licence and permissions note. Check that it runs top to bottom when opened cold.

How it is judged

The book page’s rubric governs the PDF. Three failures cap a rubric row on their own: an inbound item left with no verdict, a headline number that does not rerun from your package, and a run record older than your last edit, which counts as not reproducing. An honest “this part did not reproduce, and here is what broke” never triggers a cap. The cap is for the false pass.

Milestone 11: Your reproducible package

Open the milestone workbook in Colab

This chapter closes Studio 11: Reproduce and package. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. An independent cold-run record and a reusable research package. Working alone, the record is a clean-environment rerun labeled “solo proxy; external cold run pending”: the label, and the honesty it enforces, are part of the artifact.

What you bring

You also carry forward the last milestone’s artifact: **Milestone 10: Your artifact, ready to publish or present.** This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson’s **It is your turn** first; the practice below assumes it.

- Lesson 36: Replication and Reproduction: your cold reproduction record: every number re-run from the top, one undisclosed choice varied, and missingness classified before any filling.

- Lesson 37: Open and Reusable Research Packages: your research capsule: notebook, data or its access recipe, environment record, README, licence with its reuse boundary and restrictions, the exact output locations, and a cold rerun in a second environment.

The practice

1. Assemble the draft package first: data or its access route, code, environment, documentation, and licence.
2. Run your own package cold, from a clean environment, following only your written instructions.
3. Then the external test: have someone else run it without your help, and log every place they had to ask you a question. Working alone? Rerun it yourself after a real break, instructions only, and label the record 'solo proxy; external cold run pending': a proxy never becomes a claim of independent reproduction.
4. Fix what the runs exposed: every question asked is a missing line in your documentation, and freeze the final manifest with its date.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** What can be shared is set by your permissions, and the package says so explicitly.
- **Evidence, provenance, and reproducibility.** Reproduction is the evidence rail's final test.
- **AI activity, verification, and human decisions.** An assistant can generate documentation; only a cold human run can validate it.
- **Uncertainty, claim boundary, and revision history.** Reproduce the uncertainty statement, not only the point estimate.

Where this milestone sits

You arrived carrying Milestone 10: Your artifact, ready to publish or present. Milestone 12 decides whether the packaged work is ready to leave your hands, and what the next cycle should ask. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

A failed cold run is a finding about your work, and it precedes any release.

How this milestone is assessed

Each row scores **0, 1, or 2. 10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning

#	Criterion	0	1	2
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: An independent cold-run record (working alone: a clean-environment rerun labeled 'solo proxy; external cold run pending') and a reusable research package	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

! Blocking gate

No work at or after Studio 4 proceeds past a not-authorized determination. This is not scored and cannot be averaged away.

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.

Milestone 9: Your bounded claims

Open the milestone workbook in Colab

This chapter closes Studio 9: Write, bound, and disclose. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. A stand-alone research note or report draft, a claim-to-evidence table, and your AI disclosure statement.

What you bring

You also carry forward the last milestone's artifact: **Milestone 8: Your robustness audit**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 28: From Results to Claims: your claims with verbs downgraded to what the design established, each carrying its boundary line.
- Lesson 29: Claim–Evidence Tables: your claim-evidence table, one row per claim with claim, evidence, trace, and boundary, every number traced to its output.
- Lesson 30: AI Disclosure and Research Integrity: your AI-use disclosure, written from the ledger and naming what you did not verify.
- Lesson 31: From Dossier to Research Note: your stand-alone note draft: the problem-gap-question spine, methods, results with uncertainty, discussion, ordered findings with the headline claim's boundary inside, references, your disclosure placed, and the pointer to the package Studio 11 will freeze.

The practice

1. Write your claim as one sentence, with its boundary attached in the same sentence.
2. Build the claim-to-evidence table: every claim, the output that supports it, and the check that verified it.

3. State your uncertainty and your limitations in the words a reader outside your field would use.
4. Write your AI disclosure from your ledger, not from memory.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** Disclosure includes what you sent to tools, not only what they produced.
- **Evidence, provenance, and reproducibility.** A claim with no row in the table does not appear in the draft.
- **AI activity, verification, and human decisions.** The sentence that ships is yours, whatever drafted it.
- **Uncertainty, claim boundary, and revision history.** Findings are never communicated as certainties.

Where this milestone sits

You arrived carrying Milestone 8: Your robustness audit. Milestone 10 adapts these bounded claims to a real audience and rehearses their defense. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

A claim you cannot fit in the table is a claim to cut or a check to run, and both send you back a studio.

How this milestone is assessed

Each row scores **0, 1, or 2. 10 points total.**

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3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
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